

Clonal History and Genetic Predictors of Transformation Into Small-Cell Carcinomas From Lung Adenocarcinomas

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ABSTRACT

Purpose

Histologic transformation of *EGFR* mutant lung adenocarcinoma (LADC) into small-cell lung cancer (SCLC) has been described as one of the major resistant mechanisms for epidermal growth factor receptor (EGFR) tyrosine kinase inhibitors (TKIs). However, the molecular pathogenesis is still unclear.

Methods

We investigated 21 patients with advanced *EGFR*-mutant LADCs that were transformed into EGFR TKI-resistant SCLCs. Among them, whole genome sequencing was applied for nine tumors acquired at various time points from four patients to reconstruct their clonal evolutionary history and to detect genetic predictors for small-cell transformation. The findings were validated by immunohistochemistry in 210 lung cancer tissues.

Results

We identified that EGFR TKI-resistant LADCs and SCLCs share a common clonal origin and undergo branched evolutionary trajectories. The clonal divergence of SCLC ancestors from the LADC cells occurred before the first EGFR TKI treatments, and the complete inactivation of both *RB1* and *TP53* were observed from the early LADC stages in sequenced tumors. We extended the findings by immunohistochemistry in the early-stage LADC tissues of 75 patients treated with EGFR TKIs; inactivation of both Rb and p53 was strikingly more frequent in the small-cell-transformed group than in the nontransformed group (82% v 3%; odds ratio, 131; 95% CI, 19.9 to 859). Among patients registered in a predefined cohort (n = 65), an *EGFR* mutant LADC that harbored completely inactivated Rb and p53 had a 43× greater risk of small-cell transformation (relative risk, 42.8; 95% CI, 5.88 to 311). Branch-specific mutational signature analysis revealed that apolipoprotein B mRNA editing enzyme, catalytic polypeptide-like (APOBEC)-induced hypermutation was frequent in the branches toward small-cell transformation.

Conclusion

EGFR TKI-resistant SCLCs are branched out early from the LADC clones that harbor completely inactivated *RB1* and *TP53*. The evaluation of *RB1* and *TP53* status in EGFR TKI-treated LADCs is informative in predicting small-cell transformation.

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INTRODUCTION

Histologic transformation of *EGFR* mutant lung adenocarcinoma (LADC) into small-cell lung cancer (SCLC) has been identified as one of the major mechanisms of resistance to epidermal growth factor receptor (EGFR) tyrosine kinase inhibitors (TKIs).¹⁻³ Patients typically present with an aggressive clinical course after the transformation, which leads to a rapid deterioration of the patients'

condition.⁴ The treatment-resistant SCLCs seem to originate from preceding LADCs because they share the same type of *EGFR* mutations. However, the molecular pathogenesis of this process, especially when and how the LADC cells are transformed into SCLCs, remains elusive. A previous study indicated that the inactivation of *RB1* is common in the transformed SCLCs but is not sufficient for LADC cells to transdifferentiate into SCLCs.⁵ To understand the transformation process, we studied patients whose advanced LADCs were transformed

ASSOCIATED CONTENT



See accompanying Editorial on page 2987



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into SCLCs during EGFR TKI treatment and reconstructed their tumor clonal evolutionary history from whole genome sequencing (WGS). On the basis of these results, we examined the clinical implication of Rb and p53 expression status in early LADCs in predicting small-cell transformation.

METHODS

Study Design

This study was approved by the institutional review boards of all participating centers (Data Supplement). After approval, tissues were acquired with written informed consent. This study was conducted in observation of the most recent version of World Medical Association Declaration of Helsinki.

To identify patients with advanced LADC that was transformed into SCLC, we first screened a cohort with advanced *EGFR*-mutant LADC who underwent repetitive biopsy at the point of EGFR TKI-resistance at the Seoul National University Hospital ($n = 206$). Second, we collected tissues from patients with advanced *EGFR*-mutant LADC that transformed into EGFR TKI-resistant SCLC at other institutions ($n = 7$ from Samsung Medical Center; $n = 3$ from Seoul National University Bundang Hospital; $n = 1$ from Boramae Medical Center). Patients LC1 and LC4 received osimertinib treatment through phase I of AURA ([ClinicalTrials.gov](https://clinicaltrials.gov) identifier NCT01802632), and patient LC2 received EGF816 through a phase I/II clinical trial ([ClinicalTrials.gov](https://clinicaltrials.gov) identifier NCT02108964).

High-Throughput Sequencing and Analysis Pipeline

DNA libraries for WGS were generated by a TruSeq PCR-Free Library Preparation Kit (Illumina, San Diego, CA) from 1 μ g of genomic DNAs extracted from tumor tissues and matched normal blood samples. WGS was performed on a HiSeq X or HiSeq 2500 platform (Illumina) to generate a minimal read depth of $30\times$ for both tumors (average depth, $46.3\times$) and matched normal blood samples ($40.7\times$; Data Supplement). Process of reads alignment, calling for somatic mutations, and establishment of high-confidence substitution sets are described in the Data Supplement.

Clonal Evolution

We estimated tumor purity, ploidy, and allele-specific copy numbers from WGS data by using the Sequenza software package⁶ (Data Supplement). On the basis of this information, we calculated mutation copy number (n_{mut}) of high-confidence base substitutions from variant allele frequency information according to a previously described method.⁷ The number of chromosomal copies of each substitution (n_{chr}) was inferred from n_{mut} by selecting the closest positive integer value. Tumor cell fraction of each substitution was calculated as follows:

$$\text{Tumor cell fraction} = \frac{n_{mut}}{n_{chr}}$$

We reconstructed the phylogeny trees according to the tumor cell fraction of all the substitutions detected from serially acquired samples.⁸ The trees were validated by using the PhyloWGS software package⁹ (Data Supplement).

Decomposition of Mutational Signatures

We decomposed mutational signatures from high-confidence base substitution sets by using the deconstructSigs software package.¹⁰ Eight mutational signatures previously identified and validated in lung cancers were used for analysis (signatures 1, 2, 4, 5, 6, 13, 15, and 17).¹¹

Pathologic Examination

All pathology slides were examined by expert lung cancer pathologists (S.K., Y.K.J., and D.H.C.; Data Supplement). The result of Rb immunohistochemistry (IHC) was classified as complete absence (no nuclear expression) or wild type (normal, variable nuclear expression). The result of p53 IHC was categorized as complete absence (no nuclear expression), overexpression (intense nuclear expression), or wild type (normal, variable nuclear expression), and complete absence of expression and overexpression were interpreted as evidence of p53 inactivation as described in previous studies.^{12,13}

RESULTS

Clinicopathologic Characteristics of 21 EGFR TKI-Resistant Transformed SCLCs

We recruited 21 patients with advanced LADC that was transformed into SCLC at the point of acquired resistance to EGFR TKI-treatment (defined by Jackman criteria¹⁴). Among them, 10 patients (LC1 to LC10, which included one reported previously¹⁵ [LC1]) were from a single-institution cohort that comprised 206 patients with *EGFR* mutant-LADC who developed acquired resistance to EGFR TKIs. An additional 11 patients (LC11 to LC21) were identified from the medical oncology clinics of three different institutions (clinical history and pathologic findings of five patients were previously described in the literature^{16,17}). All 21 patients were initially given a diagnosis of an advanced LADC with canonical *EGFR*-activating mutation and underwent EGFR TKI-treatment to achieve a favorable response (Table 1). Among them, four (19%) had *EGFR* L858R mutation and six (29%) were male; both characteristics were rarely observed in the literature on small-cell transformation.³ All but two patients (90%) were never-smokers or light smokers (< 10 pack-years). By using microscopic examination and IHC, we confirmed histology of all available serially acquired tumors.

Early Branched Ancestry of Transformed SCLC From the LADC Lineage

To explore the relationship between transformed SCLCs and their counterpart, early LADCs, we performed WGS of serially acquired tumors. These longitudinal sequencing analyses were available for four (LC1 to LC4) of the 21 patients. Overall, nine tumors (five SCLCs and their corresponding early LADCs; all fresh frozen except for two formalin-fixed paraffin-embedded early LADC tissues [LC2b and LC3a]) were sequenced for whole genomes, and their treatment history is summarized in Figure 1 and the Data Supplement. In addition, we sequenced an SCLC tissue from a male light smoker that harbored an *EGFR* L858R mutation (LC0b), which was primarily resistant to gefitinib. We identified high-confidence somatic mutations, including base substitutions (Data Supplement), short indels, and structural variations. Alterations in known cancer genes are summarized in the Data Supplement.

We first used the high-confidence somatic base substitutions to investigate the clonal relationship between paired tumors of each patient. Two-dimensional plotting of tumor cell fractions, the fraction of tumor cells that harbor each substitution, identified similar distributions with three mutational clusters in all patients

Table 1. Clinicopathologic Characteristics of Patients With Advanced *EGFR* Mutant LADC Transformed Into SCLC

Patient ID	Sex	Age*	Active Smoking	Initial Histology		Transformed Histology		Treatment Before Transformation
				Histology	EGFR Mutation	Histology	Neuroendocrine Markers (C/S/N)	
LC1	F	56	Never	LADC	Del E746-A750	SCLC	+/+	Chemotherapy, gefitinib, osimertinib
LC2	F	39	Never	LADC	Del E746-A750	SCLC with spindle cells	+/+	Gefitinib, chemotherapy, EGF816
LC3	F	51	Ex-smoker, 0.5 pack-y	LADC	L858R	LADC + SCLC	+/+	Gefitinib
LC4	F	39	Ex-smoker, 1.5 pack-y	LADC	Exon 19 del (not specified)	SCLC	+/-	Osimertinib
LC5	M	50	Never	LADC	Del E746-T751 ins VP	SCLC	-/+	Dacomitinib, chemotherapy, gefitinib
LC6	F	79	Never	LADC	Del E746-A750	LADC + SCLC	+/+	Gefitinib
LC7	F	61	Never	LADC	Del E746-A750	SCLC	-/+	Dacomitinib
LC8	F	61	Never	LADC	L858R	LADC + SCLC	+/+	Chemotherapy, gefitinib
LC9	F	58	Never	LADC	Del L747-P753	SCLC	+/-	Gefitinib
LC10	M	62	Never	LADC	Del L747-T751	SCLC	+/-	Gefitinib
LC11	F	54	Never	LADC	Del L747-A755 ins SRD	LADC + SCLC	ne/ne/ne	Chemotherapy, afatinib
LC12	F	55	Never	LADC	Exon 19 del (not specified)	SCLC	ne/ne/+	Gefitinib, osimertinib
LC13	F	59	Never	LADC	Exon 19 del (not specified)	LADC + SCLC	ne/ne/-	Chemotherapy, gefitinib, rociletinib
LC14	M	37	Ex-smoker, 1 pack-y	LADC	Exon 19 del (not specified)	SCLC	+/+	Gefitinib, chemotherapy, osimertinib
LC15	F	57	Never	LADC	L858R	SCLC	ne/ne/+	Erlotinib, chemotherapy, osimertinib
LC16	F	58	Never	LADC	Exon 19 del (not specified)	SCLC	ne/ne/+	Chemotherapy, erlotinib, afatinib, osimertinib
LC17	M	53	Ex-smoker, 5 pack-y	LADC	Exon 19 del (not specified)	SCLC	ne/ne/+	Erlotinib, chemotherapy, afatinib
LC18	F	56	Never	LADC	Del E746-A750	SCLC	-/+	Afatinib
LC19	F	64	Never	LADC	Del E746-A750	SCLC	ne/+	Chemotherapy, gefitinib, osimertinib
LC20	M	66	Ex-smoker, 40 pack-y	LADC	L858R	SCLC	ne/+	Erlotinib
LC21	M	43	Ex-smoker, 18 pack-y	LADC	Del E746-A750	SCLC	ne/+	Erlotinib

Abbreviations: C/S/N, chromogranin A/synaptophysin/neural cell adhesion molecule; EGFR, epidermal growth factor receptor; LADC, lung adenocarcinoma; ne, not examined; SCLC, small-cell lung cancer.

*Age at the diagnosis of stage IV disease.

(Fig 2A; Data Supplement). The first cluster was observed at [1.0, 1.0], which indicated truncal mutations were present in the most recent common ancestor cell of early LADC and transformed SCLC clones. The first clusters in two patients (Fig 2A) commonly had EGFR-activating mutations and inactivating mutations of *RB1* and *TP53*. The second and the third clusters (present on the x - and y -axis, respectively) included late mutations on the branches, which were private for early LADCs and transformed SCLCs, respectively. Of note, this branched clonal relationship implies that the transformed SCLC clones did not originate directly from the major clones of preceding LADCs in all four patients. We further validated this exclusive clonal relationship by deeper sequencing of tumors from patient LC1 (LC1b and LC1c, exomes with approximately 150 $\times$ coverage and targeted deep sequencing with approximately 20,000 $\times$ coverage; Data Supplement).

We further investigated the time point of clonal divergence. In patients LC1 and LC2, the early LADCs (LC1b and LC2b) were gefitinib-resistant-LADCs, and the transformed SCLCs (LC1c and LC2c) were diagnosed during subsequent treatments with third-generation EGFR TKIs. Of note, secondary *EGFR* alterations that conferred gefitinib resistance (ie, *EGFR* T790M^{18,19} and focal *EGFR* amplification) were only present in the gefitinib-

resistant-LADCs (LC1b and LC2b) and not in their counterpart SCLCs (LC1c/d [Fig 2] and LC2c [Data Supplement]). In contrast, some mutations, including *PIK3CA* mutations and *MYC* amplifications, were only observed in the transformed SCLCs, which implies that the third-generation EGFR TKI-resistant SCLC clones (major clones of LC1c and LC2c) were branched out before the acquisition of the focal *EGFR* amplification and/or T790M substitutions in their counterpart LADCs (major clones of LC1b and LC2b). This finding suggests that the clonal divergence likely preceded gefitinib treatment in both patients, although overt small-cell transformation appeared in the late stages (ie, 25 to 30 months after initial diagnosis). Furthermore, the branched clonal relationship in patient LC3 (Data Supplement) and patient LC4 (Fig 2) indicated that the branch out of SCLC ancestor clones even preceded the time points of initial diagnosis. In patient LC4, a retention of heterozygosity of chromosome 22 was observed in the transformed SCLC (LC4b), whereas a loss of heterozygosity was evident in early LADC (LC4a; Fig 2B); this strongly supports the branched evolutionary relationship between LC4a and LC4b. Clonal comparison in patient LC3 was more challenging because of the low purity of LC3a and mixed histology of LC3b (LADC + SCLC); however, many high-confidence substitutions only present in LC3a (n = 453) also

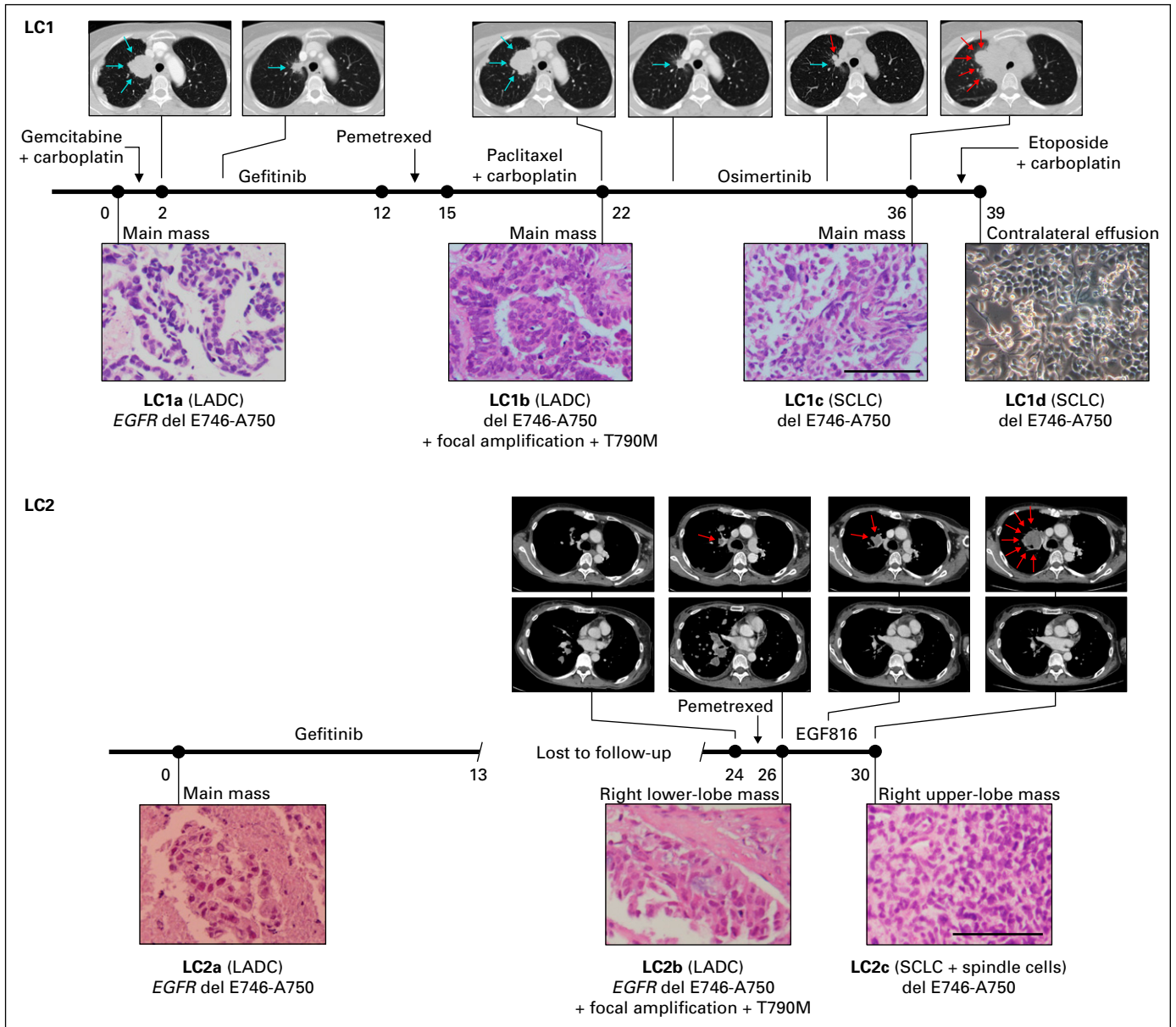


Fig 1. Clinical history of two patients who underwent whole genome sequencing (WGS) with transformed small-cell lung cancers (SCLCs). Treatment history of two patients who underwent WGS and points of tissue acquisitions are described in timelines. Numbers indicate time (in months) from the diagnosis of advanced lung adenocarcinoma (LADC). Scale bar indicates 100 μ m. The blue arrows in computed tomography images indicate LADC masses, and the red arrows indicate SCLC masses.

supported a branched clonal relationship between the paired tumors. We summarize the history of clonal evolution in four sequenced patients in Figure 3A.

Small-Cell Transformation Is Predictable by Early Complete Inactivation of *RB1* and *TP53*

We found that the biallelic inactivation of both *RB1* and *TP53* was observed from the early LADC stages in all the sequenced tumors (Fig 3B) except for one allele of *TP53* in LC3a presumably because of the low purity or tumor heterogeneity of the sample (Data Supplement). At the molecular level, the inactivation of one parental copy of these genes was caused by diverse mechanisms,

including frameshift indels, large deletions, and nonsense or inactivating missense substitutions. The other parental copy was invariably inactivated by whole chromosomal- or chromosomal arm-level loss-of-heterozygosity events in all four patients (Fig 3B; Data Supplement). These biallelic inactivation events in both *RB1* and *TP53* were consistently shared by counterpart SCLCs in each patient. Although the biallelic, complete inactivation of the two canonical tumor suppressor genes are genetic hallmarks of the primary SCLCs,²⁰ it is unusual for ordinary LADCs.²¹ Our re-analysis of The Cancer Genome Atlas data set indicated that 20 (5%) of the 416 patients with LADCs had mutations in both *RB1* and *TP53*,²¹ the prevalence of which agrees with the observed prevalence of small-cell transformation (5%) in the current EGFR

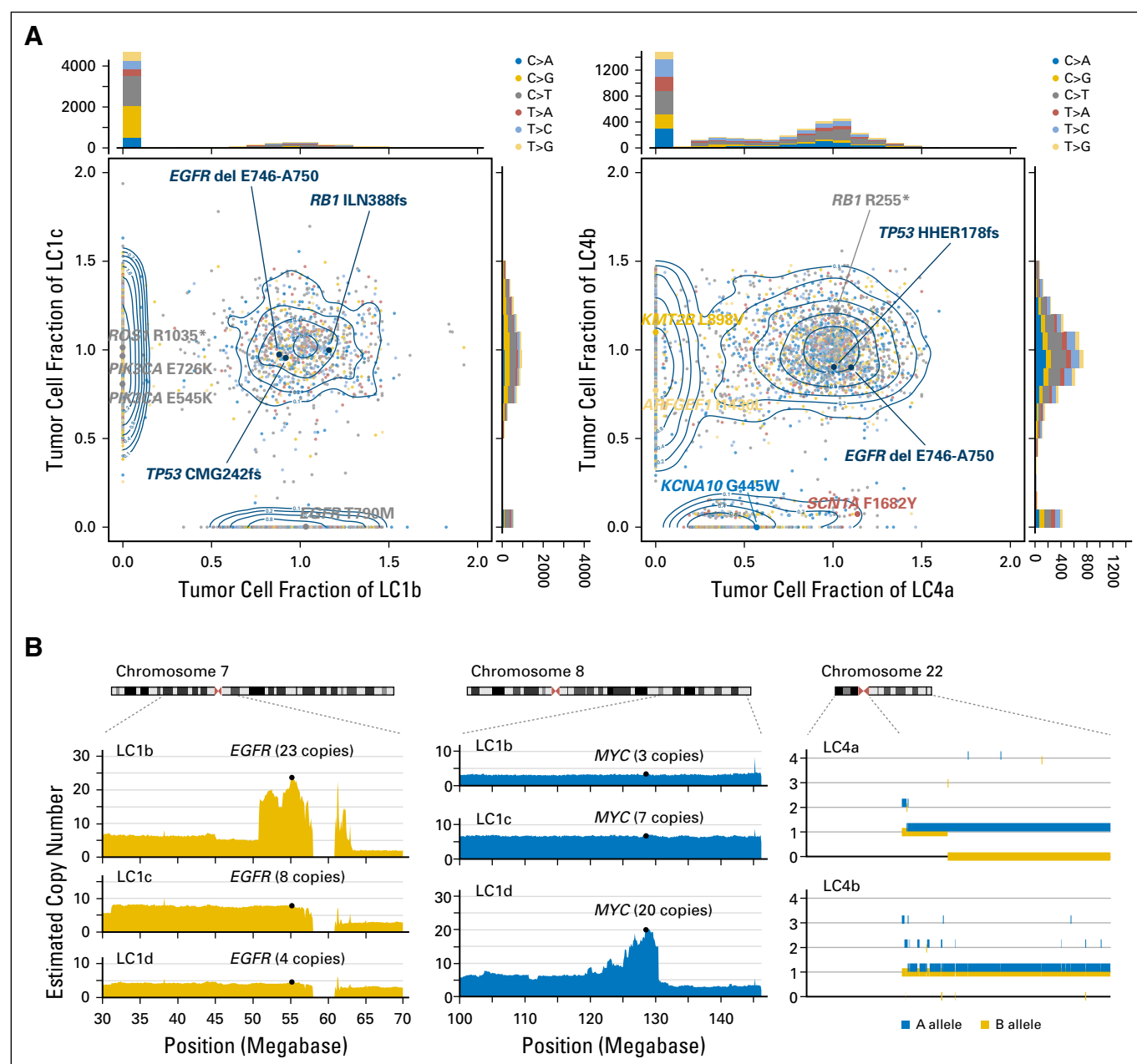


Fig 2. Branched clonal evolution of the transformed small-cell lung cancers (SCLCs). (A) Clonal fractions of somatic base substitutions (tumor cell fraction) were compared between early lung adenocarcinomas (LADCs; LC1b and LC4a) and epidermal growth factor receptor tyrosine kinase inhibitor-resistant SCLCs (LC1c and LC4b). LC1b is a gefitinib-resistant *EGFR* T790M-positive LADC, and LC4a is a treatment-naïve initial LADC. In each plot, three clusters of mutations are identified (demarcated with blue contours). Mutations in cancer-related genes are indicated. (B) Smoothed copy numbers of LC1 tumors are plotted against the genomic positions of chromosome 7 (in which the *EGFR* is located) and 8 (*MYC*). Allele-specific copy number estimates of LC4 tumors are plotted for chromosome 22.

TKI-acquired resistance cohort (Data Supplement) and within the range of what was reported for SCLC transformation in the literature.^{1,2} Therefore, we speculated that LADCs with the complete inactivation of both *RB1* and *TP53* are predisposed to transform into SCLCs. To elucidate this idea further, we evaluated the status of Rb and p53 by IHC in a wide range of 210 lung cancer tissues (n = 210), including those that were small-cell transformed.

First, we confirmed the expression of Rb and p53 in typical primary LADCs by using tissue arrays (n = 121; Fig 4A). Among

121 surgically resected LADCs, the inactivation of Rb and p53 was observed in five (4%) and 36 (30%), respectively. Overall, three tissues (2%) had concomitant inactivation of both Rb and p53. These data demonstrate the validity of IHC in assessing the status of Rb and p53 in LADC tissues.

Next, we extended the IHC analysis to the initial *EGFR* mutant LADC tissues from patients with stage IV disease who received *EGFR* TKI treatment and ultimately developed a resistant tumor with a confirmed histology (Fig 4B). In a case-control comparison

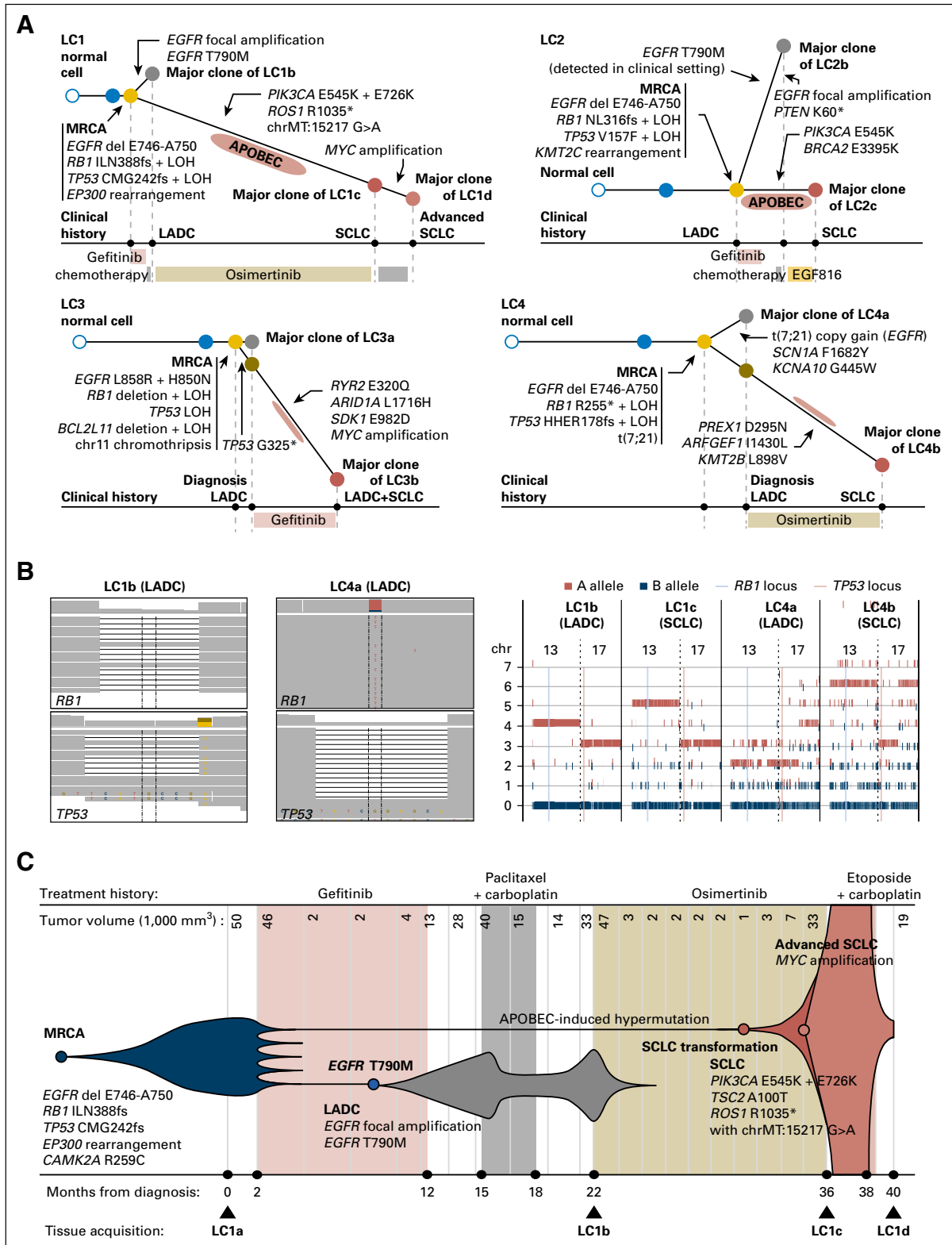


Fig 3. Phylogeny analysis of serially acquired tumors. (A) Phylogeny trees of sequenced patients (LC1 to LC4) are reconstructed from the whole genome sequencing data. Circles indicate major clones of the tumors. The length of each branch is proportional to the number of mutations that occurred in the branch. Mutations of cancer-related genes in each branch are indicated with arrows. The time points of relevant treatments are summarized beneath the trees. (B) Mutations of *RB1* and *TP53* in two early lung adenocarcinomas (LADCs; LC1b and LC4a) are visualized by using Integrative Genomics Viewer (left panel). Allele-specific copy number analysis shows loss of heterozygosity of chromosomes 13 and 17 in both early LADCs and epidermal growth factor receptor tyrosine kinase inhibitor-resistant small-cell lung cancers (SCLCs; right panel). (C) Clonal evolution of LC1 is described with clinical history and tumor volumes. The horizontal axis represents the time from the diagnosis (0), and the vertical axis indicates the volume of tumors calculated from the computed tomography images. MRCA, most recent common ancestor.

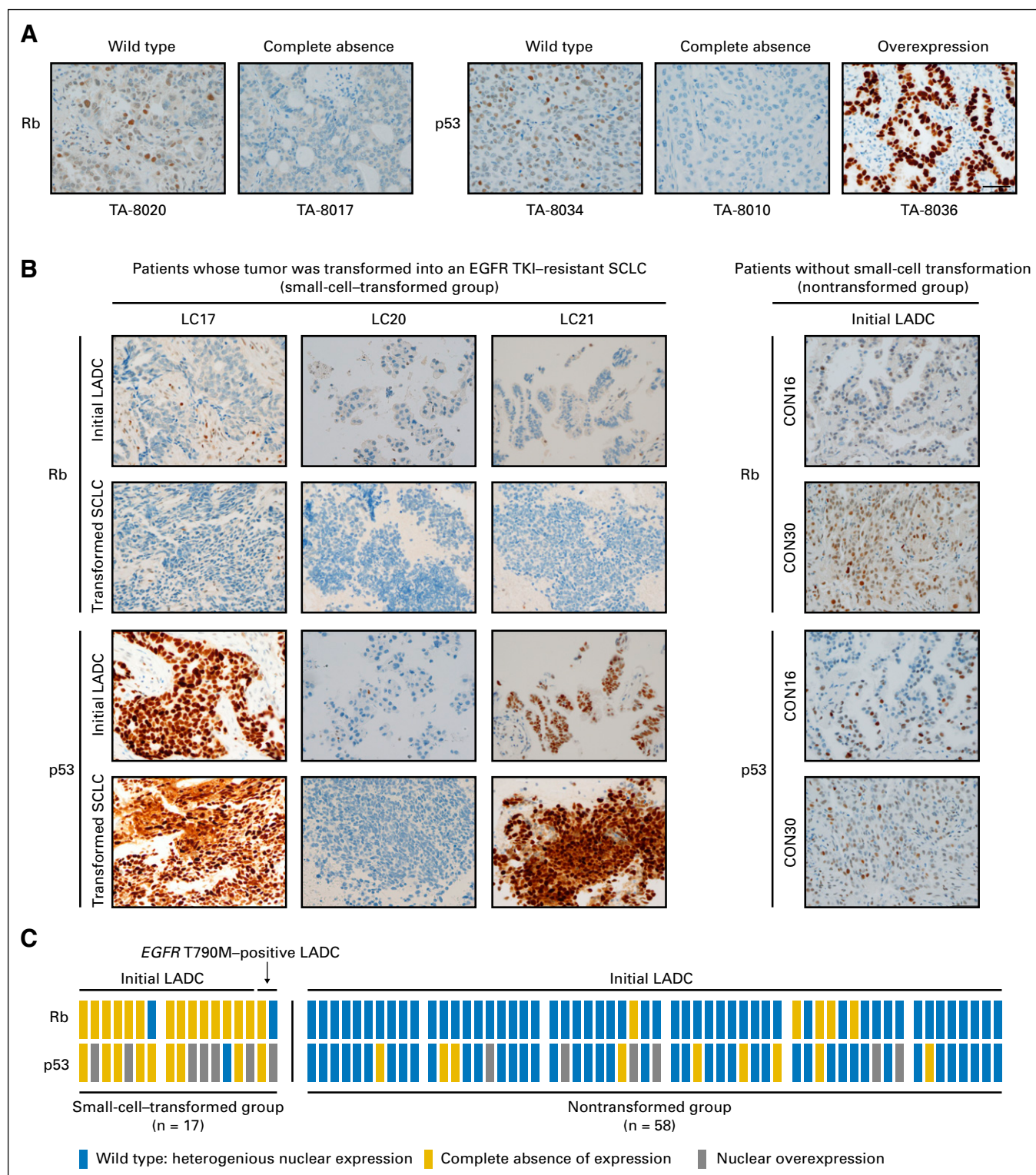


Fig 4. Inactivation of Rb and p53 in early lung adenocarcinomas (LADCs) and their relationship with small-cell transformation. (A) Immunohistochemical staining of Rb and p53 in surgically resected early-stage LADCs. Representative images are shown (identification keys in the tissue array indicated beneath the images). Scale bar on the TA-8036 image indicates 50 μ m, and all images were taken under $\times 400$ magnification. (B) Expression status of Rb and p53 in tumor tissues of patients in the small-cell-transformed group (LC17, LC20, and LC21) and nontransformed group (CON16 and CON40). In the small-cell-transformed group, all tumors, including initial LADCs and transformed small-cell lung cancers (SCLCs), showed no expression of Rb. For p53, tumor pairs of LC17 and LC21 showed intense overexpression, whereas tumor pairs of LC20 showed complete absence of p53 expression. In contrast, the initial LADC tissues of CON16 and CON30 showed wild-type expression of Rb and p53. In CON16 and CON30, small-cell transformation was not observed during their clinical history, and their tumors became resistant to epidermal growth factor receptor (EGFR) tyrosine kinase inhibitors (TKIs) by acquisition of *EGFR* T790M mutation. (C) Summary of the result of Rb and p53 immunohistochemical staining.

between the small-cell-transformed ($n = 17$) and non-transformed ($n = 58$) groups, we found that the complete inactivation of Rb and p53 was enriched among the initial LADCs of the small-cell-transformed group (14 of 17 v two of 58; odds ratio, 131; 95% CI, 19.9 to 859; Fig 4C). In three patients (18%), either gene was intact in the initial LADCs ($n = 3$) but subsequently inactivated in the transformed SCLC tissues. Except for the complete inactivation of both Rb and p53, we identified no other significant difference in clinicopathologic variables between the small-cell-transformed and nontransformed groups (Table 2). When we focus on the 65 patients ($n = 7$ small-cell-transformed; $n = 58$ nontransformed) in the predefined acquired resistance cohort from Seoul National University Hospital, LADCs with completely inactivated Rb and p53 had a 43× greater likelihood to transform into SCLCs during treatment (relative risk, 42.8; 95% CI, 5.88 to 311). These data demonstrate that complete inactivation of both *RB1* and *TP53* in LADC is a prerequisite for small-cell transformation and is detectable by IHC early during the LADC stage and, therefore, could be used as a strong predictive biomarker in clinics.

Apolipoprotein B mRNA Editing Enzyme, Catalytic Polypeptide-Like–Induced Hypermethylation in the Branch of Small-Cell Transformation

To characterize the genomic instability process during small-cell transformation, we investigated mutational signatures.²² The apolipoprotein B mRNA editing enzyme, catalytic polypeptide-like (APOBEC)–associated mutational signatures (2 and 13) were

preferentially observed in the mutational spectrum of SCLC branches in patients LC1 and LC2 (56% and 43% of mutations in these branches, respectively) compared with their gefitinib-resistant LADC branches (18% in LC1 and 14% in LC2; Data Supplement). In contrast, the APOBEC-associated mutational signatures were less frequent (10% to 11%) in evolutionary branches from initial LADCs to transformed SCLCs in patients LC3 and LC4. Of note, our reanalysis of exome sequencing data from an autopsy⁵ also demonstrated preferential APOBEC-induced hypermutation in transformed SCLCs (30% to 33%), whereas *EGFR* T790M–positive LADC had rare APOBEC-associated mutations (5%). The APOBEC mutational signature also was prevalent in the patient with primary resistance (LC0b; 40% of all substitutions; Data Supplement). The APOBEC mutational process is known to frequently activated in LADCs and may drive subclonal diversification.^{23,24} The current analysis could provide further evidence that the APOBEC-induced mutational process can be hyperactivated during transformation into SCLCs.

Finally, we integrated tumor volume information, which was calculated from computed tomography images acquired through the treatment history of patient LC1 into our clonal analysis. As described in Figure 3C, the transformed SCLC and gefitinib-resistant LADC in patient LC1 underwent early branched and parallel evolutionary trajectories. The clonal divergence seemed to occur before the gefitinib treatment, and *RB1* and *TP53* were already inactivated at this point. A long latent period before the small-cell transformation, in which the APOBEC-induced hypermutation was operative, was apparent.

Table 2. Factors Associated With Histologic Transformation Into SCLC

Factor	No. of Patients	Transformation Into SCLC, No. (%)	Odds Ratio	95% CI
Age, years				
< 65	52	15 (28.9)	Reference	
≥ 65	23	2 (8.7)	0.235	0.0489 to 1.13
Sex				
Male	25	5 (20.0)	Reference	
Female	50	12 (24.0)	1.26	0.390 to 4.09
Smoking				
Never-smoker	54	12 (22.2)	Reference	
Ever-smoker	21	5 (23.8)	1.09	0.332 to 3.60
<i>EGFR</i> mutation				
L858R	30	4 (13.3)	Reference	
Exon 19 del	43	13 (30.2)	2.82	0.817 to 9.71
Use of third-generation <i>EGFR</i> TKIs*				
No	35	9 (25.7)	Reference	
Yes	40	8 (20)	0.722	0.244 to 2.14
Time from diagnosis to last <i>EGFR</i> TKI failure, months†				
< 30	41	10 (24.4)	Reference	
≥ 30	34	7 (20.6)	0.804	0.269 to 2.40
Rb and p53 status in preceding LADC tissue				
Intact either	59	3 (5.1)	Reference	
Both inactivated	16	14 (87.5)	131	19.9 to 859

NOTE. Reference = 1.

Abbreviations: *EGFR*, epidermal growth factor receptor; LADC, lung adenocarcinoma; SCLC, small-cell lung cancer; TKI, tyrosine kinase inhibitor.

*Third-generation *EGFR* TKIs included osimertinib, rociletinib, olmutinib, and EGF816.

†In all patients whose tumor underwent small-cell transformation, disease progression events to the last *EGFR* TKI treatment were attributable to small-cell transformation.

DISCUSSION

Overall, this study indicates that EGFR TKI-resistant SCLC clones can be derived from divergent evolutionary processes from LADCs at the very early stage. Most importantly, this transformation process is already predisposed at the stage of early LADCs by complete inactivation of Rb and p53, which are predictive biomarkers for small-cell transformation.

Although the majority of transformed SCLC cases have been identified from EGFR-mutant LADC cases, studies have indicated that this phenomenon is not restricted to specific molecular subtypes; small-cell transformation could also occur in ALK-rearranged LADCs during ALK inhibition²⁵ or at recurrence after surgical resection of a driver-unknown LADC.²⁶ We also examined the expression of Rb and p53 in an initial LADC tissue without EGFR mutation from a previously reported patient¹⁶ whose tumor underwent transformation into SCLC during treatment. In the initial LADC as well as in the transformed SCLC, Rb and p53 were also completely inactivated. These findings suggest that small-cell transformation is more likely a clonal evolution event in LADC cells that is based on complete inactivation of *RB1* and *TP53*. Therefore, an examination of the status of Rb and p53 in advanced LADCs merits consideration; LADCs with inactivation of these two genes may warrant observations for transformation into SCLC during their clinical history.

Multiplex genotyping of targetable cancer-related genes is becoming popular in medical oncology clinics, especially in those that treat patients with lung cancer.²⁷ However, for tumor suppressors such as *RB1* and *TP53* in which the mutations are distributed along the entire length of a gene rather than concentrated in hot spots, the detection of all mutations sensitively by targeted deep sequencing is difficult. For example, *RB1* is frequently disrupted by genomic rearrangements of which breakpoints are frequently located in introns²⁰; therefore, the detection of *RB1* mutations sensitively by targeted deep sequencing assays is difficult. In this regard, we believe that the IHC for Rb is a reliable alternative assay for the *RB1* mutation analysis. The IHC for p53 also showed a good correlation between mutational data in previous studies^{12,13} and the current study.

Our patients who underwent WGS showed favorable and prolonged response to EGFR TKIs. For example, patient LC1 exhibited −96.5% and −96.9% of tumor volume shrinkage in response to gefitinib and osimertinib, respectively, and those responses were durable for > 10 months. Patient LC4 also showed −95.9% of tumor volume reduction in response to osimertinib, which lasted for > 2 years. However, how the direct ancestor cells of transformed SCLCs, which were likely to exist as an LADC subclone before overt SCLC transformation, survived for approximately 30 months in the presence of effective EGFR inhibition without acquiring any known drug-resistant mutations (ie, EGFR T790M, C797 substitutions,²⁸ or *MET* amplifications²⁹) is unknown. Presumably, a small number of full- or semi-transformed, thus drug-resistant, SCLC clones could be present at some earlier LADC stage. Otherwise, a single LADC clone should somehow tolerate the long period without acquiring drug-resistant mutations. In either scenario, a few cancer cells should be dormant for a long period before overt clinical transformation, which can be explained by the drug persister concept that refers to a small subset

of slow-proliferating cells in the tumor tissue resistant to apoptosis.³⁰⁻³² Of note, the number of C>T transitions in the CpG context (designated as signature 1, which originates from spontaneous deamination of methylcytosines), which are known to be proportional to the number of mitoses,³³ among the transformed SCLC-private mutations was smaller than that among the LADC-private mutations in both patients LC1 and LC2, despite the SCLCs acquired at later time points than the LADCs (Data Supplement). Thus, cell division time might be longer in the SCLC branch, which resembles the slow-growing properties of persister cells.³¹ During this stage, the APOBEC-induced hypermutation could occur in these latent cancer cells and is reminiscent of stress-induced hypermutation and related phenotypic switch in microbes during antimicrobial treatment, which maximizes their adaptability against selection pressure from the drugs.³⁴

This study was initiated to understand the clonal history of EGFR TKI-resistant SCLCs transformed from LADCs by using longitudinal sequencing. Although a pattern of early branched evolution is suggested in all four patients, the small sample size does not exclude the possibility of other evolutionary trajectories (ie, linear evolution). For a more complete understanding, a similar study with a larger number of patients who undergo WGS may be necessary. In particular, the acquisition of high-quality paired tumor samples, including primary tumors, is challenging in clinic, and we have carefully included data from two formalin-fixed paraffin-embedded primary tumors and performed a conservative analysis for valid clonal comparisons (Data Supplement) that may be partly offset by the exclusion of true-positive mutations from the analysis.

This study highlights the common clonal dynamics and mutational processes that actively operate in LADC cells from the very early stage. An understanding of the patterns of cancer clonal evolution is essential for designing an optimal treatment strategy.

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Disclosures provided by the authors are available with this article at jco.org.

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Clonal History and Genetic Predictors of Transformation Into Small-Cell Carcinomas From Lung Adenocarcinomas

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